

Beyond the Scoreline — Final Outcome Report

Final_hypo/ — built and executed against *Beyond the Scoreline.docx*, on a half-time dataset rebuilt from StatsBomb (Hudl) open event data specifically for this study.

This document reports what the proposal asked for, against what was actually built and measured. Every number below is read directly off the result CSVs in *results/* or the executed notebook *notebooks/beyond_the_scoreline_final.ipynb* (52 cells, 0 execution errors) — nothing here is estimated or reconstructed from memory.

0. Why a new dataset had to be built

The proposal's core requirement — half-time performance indicators (possession, passing, tackles, interceptions, clearances, saves, shot placement) split **strictly at the 45-minute mark** — does not exist in any conventional match-results dataset. football-data.co.uk (used throughout the rest of the project) carries full-match shot/card totals only, with no half-time split of any performance statistic and no possession or passing data at all.

Solution. StatsBomb open data publishes every on-ball event of a match with a *period* field, so filtering to *period == 1* gives an *exact* half-time cut. *Final_hypo/src/ingest.py* downloads and aggregates this directly from source (not from the two *league_stats.csv/knockout_stats.csv* files supplied partway through the project, which were missing the target variable, dates, and match-weeks needed for the proposal's required temporal validation).

Scope. The only block of StatsBomb's archive with complete, comparable, multi-league coverage of a single season: the **Big-4 leagues, 2015/16** (Premier League, La Liga, Ligue 1, Serie A) — **1,517 matches**, all fully ingested. A secondary set of **250 tournament matches** (FIFA World Cup, UEFA Euro, UEFA Women's Euro — group and knockout stages) was ingested for the format-comparison extension in §9.

1. Dataset

Matches	1,517 (Premier League 380, La Liga 380, Serie A 380, Ligue 1 377)
Season	2015/2016, match-weeks 1–38
Target	Full-time result, H / D / A
Class balance	H 44.5%, A 29.0%, D 26.5%
Development block	1,278 matches (weeks < 33)
Frozen holdout	239 matches (weeks 33+), untouched by any development step
Performance indicators	21 first-half difference features: possession, shots, shots on target, shots off target, blocked shots, shots inside box, passes attempted, passes completed, pass accuracy,

corners, free kicks, tackles, tackles won, interceptions, clearances, saves, fouls, yellow cards, red cards, offsides

Feature architecture, exactly as specified (proposal §1–§4): match perspective with difference features ($X_{diff} = X_{home} - X_{away}$), not per-team rows — this is what stops a model from learning "this club tends to win" instead of "the team controlling the half tends to win". `HT_state` is excluded from the models as deterministic (proposal §2).

Set	Contents	Features
A	half-time scoreline only	3
B	performance indicators only, no score	21
C	A ∪ B	24

The half-time-state story, before any model is fitted (`matches_features.csv`):

Half-time state	n	→ Home win	→ Draw	→ Away win
Home leading	515	79.6%	14.4%	6.0%
Level	648	36.6%	39.5%	23.9%
Home trailing	354	7.9%	20.3%	71.8%

2. Methodology — what the proposal specified vs. what was built

Proposal requirement	Status
Match-perspective, diff features	✓ built (§1 above)
Models A / B / C	✓ built
6 algorithms (LR, NB, DT, RF, SVM, kNN) × 3 sets = 18 configs	✓ built
Time-aware, expanding-window CV	✓ built — by match-week , not by season (StatsBomb has no multi-season complete coverage for any league; see §11 Limitations)
Nested CV for hyperparameter tuning	✓ built — inner <code>StratifiedKFold</code> inside every training window
Majority-class baseline	✓ built, plus a class-frequency baseline and a half-time-leader rule
Full metric suite (accuracy, macro P/R/F1, weighted F1, per-class P/R/F1, log loss, macro ROC-AUC, Brier)	✓ all computed, every fold
H1–H5 as separately defined hypotheses	✓ all five run

Proposal requirement	Status
McNemar's test for H3	✓ implemented directly (binomial/chi-square form), used for the central comparison
Permutation importance and leave-one-feature-out ablation for H4	✓ both, on every fold
Score + single-feature incremental sweep for H5	✓ built, all 21 indicators
Fold-level stability (§13)	✓ built into every permutation/ablation/incremental result
H5 by match state (§14 extension)	✓ built
Frozen final holdout, untouched during development (§20)	✓ built — 239 matches, evaluated once

3. H1 — Is the half-time scoreline a strong predictor?

Hypothesis: the half-time scoreline is a strong predictor of the final match outcome.

Model A (3 features) across all 6 algorithms, expanding-window CV:

Algorithm	Accuracy	Macro F1	Weighted F1	Log loss	ROC-AUC	Brier
NaiveBayes	0.6105	0.6073	0.6234	0.919	0.775	0.504
DecisionTree	0.6071	0.5894	0.6074	0.987	0.771	0.494
kNN	0.5910	0.5639	0.5886	0.976	0.759	0.508
SVM	0.5970	0.5548	0.5786	0.890	0.732	0.524
RandomForest	0.5946	0.5519	0.5760	0.925	0.769	0.496
LogisticRegression	0.6060	0.5374	0.5676	0.836	0.777	0.491

Baselines: majority class 0.2020 macro F1, class-frequency 0.2020, **half-time-leader rule 0.6073** — exactly tying the best trained model.

Verdict: H1 SUPPORTED. All 6 algorithms clear the majority baseline by a wide margin (macro F1 +0.34 to +0.41 over 0.2020), consistently across every temporal fold. The scoreline is not just informative — for this data, a model adds essentially nothing over the one-line rule "whoever leads at half-time wins, otherwise draw."

4. H2 — Is there predictive signal in performance indicators alone?

Hypothesis: half-time performance indicators alone can predict match outcomes with reasonable accuracy.

Model B (21 features, zero score information):

Algorithm	Accuracy	Macro F1
LogisticRegression	0.5821	0.5219
SVM	0.5559	0.5095
NaiveBayes	0.4887	0.4714
RandomForest	0.5273	0.4469
DecisionTree	0.4806	0.4377
kNN	0.4544	0.4165

Verdict: H2 PARTIALLY SUPPORTED. Best Model B (0.5219) clears the baseline (0.2020) comfortably — performance indicators do carry real signal. But it trails the best Model A (0.6073) by **0.0854 macro F1**, despite having seven times as many features. Twenty-one measurements of *how* the half was played still lose to three numbers describing *what it was worth*.

5. H3 — Does performance add anything beyond the scoreline?

Hypothesis: combining half-time score and performance indicators yields significantly better predictions than the scoreline alone. The proposal specifies McNemar's test on paired match-level errors for this comparison.

Model C (A ∪ B) vs Model A, per algorithm:

Algorithm	A macro F1	C macro F1	Δ	McNemar p
DecisionTree	0.5894	0.5710	−0.0184	0.597
NaiveBayes	0.6073	0.5863	−0.0210	0.499
RandomForest	0.5519	0.5595	+0.0076	0.679
SVM	0.5548	0.5562	+0.0014	1.000
kNN	0.5639	0.5202	−0.0437	0.311
LogisticRegression	0.5374	0.5363	−0.0011	0.659

Mean Δ macro F1 across all 6 algorithms: **−0.0125**. McNemar significant ($p < 0.05$) for **0 of 6** algorithms.

Confirmed on the frozen holdout (239 matches, never touched during development): best Model A (NaiveBayes) = 0.6100, best Model C (DecisionTree) = 0.5965 — Model C still trails, and still ties the half-time-leader rule (0.6100) rather than beating it.

Verdict: H3 NOT SUPPORTED. This is the proposal's central hypothesis, and the direct answer is no — combining the scoreline with performance indicators does not produce a statistically significant improvement, on the CV protocol or on an independent frozen holdout. This is a finding, not a failure: it is the literal answer to what the title asks.

6. H4 — Which indicators contribute almost nothing?

Two independent measures, as the proposal specifies, both on the best-performing Model C algorithm (NaiveBayes, macro F1 0.5863):

Permutation importance (top and bottom of 24 features):

Feature	Importance
HT_goal_difference	0.0617
HT_away_goals	0.0448
HT_home_goals	0.0383
shot_accuracy_diff	0.0137
shots_on_target_diff	0.0122
...	...
saves_diff	−0.0012
fouls_diff	−0.0011

Leave-one-feature-out ablation (Δ macro F1 from removing the feature; the three scoreline features dominate, as expected):

Feature	Δ macro F1
HT_goal_difference	0.0456
HT_home_goals	0.0350
HT_away_goals	0.0210
red_cards_diff	0.0095
shot_accuracy_diff	0.0088
...	...
possession_diff	−0.0063
passes_completed_diff	−0.0092

Agreement between the two measures across all 24 features: **$r = 0.918$** .

Verdict: H4 SUPPORTED. Two indicators (*saves_diff*, *fouls_diff*) are flagged low-value by *both* measures — everything else that scores low on one measure is at least mildly positive on the other, which is itself informative: the disagreement is concentrated in a genuinely small, identifiable set. The picture the two measures agree on is unambiguous: **the three scoreline features account for essentially all of Model C's predictive power**; every performance indicator is an order of magnitude smaller.

7. H5 — Which indicators retain predictive value above the scoreline?

For every one of the 21 performance indicators, *Score + indicator* is compared against *Score* alone:

Feature	Δ macro F1	Folds improved (of 11)
shots_on_target_diff	+0.0095	8
pass_accuracy_diff	+0.0068	6
corners_diff	+0.0057	6
possession_diff	+0.0054	5
shots_diff	+0.0048	7
passes_completed_diff	+0.0046	5
...	...	...
shot_accuracy_diff	-0.0118	3

The best performer, **shots_on_target_diff**, improves on 8 of 11 folds — but its 95% confidence interval (± 0.0122) does not clear zero ($0.0095 - 0.0122 < 0$), and no other indicator comes close.

Verdict: H5 NOT SUPPORTED. Zero of 21 indicators survive the bar of (a) a positive mean improvement, (b) a majority of folds improved, and (c) a 95% CI that excludes zero. **shots_on_target_diff** is the nearest thing to a survivor — consistent with every other dataset in this project's history, where shots on target is repeatedly the strongest performance indicator — but it does not clear the bar here.

Cross-check against H4. The two hypotheses are consistent: nothing in H4's ablation stood out as strongly load-bearing beyond the scoreline, and nothing in H5's incremental sweep found an indicator that reliably adds value. The project's title question — "is there information beyond the scoreline?" — gets a clean, matching no from both independent angles, on this dataset.

8. §14 Extension — does an indicator's value depend on match state?

The same H5 incremental sweep, evaluated separately within each half-time state (Leading / Drawing / Trailing), using the *same* fitted folds:

State	Top feature	Δ macro F1	Folds improved
Leading	(no separation — all deltas 0.0000)	0.0000	0/11
Drawing	shots_on_target_diff	+0.1446	11/11
Drawing	shots_diff	+0.1421	10/11
Trailing	(no separation — all deltas 0.0000)	0.0000	0/11

This is the sharpest secondary finding in the whole study. When a match is level at half-time — the genuinely uncertain case, and the one the transition table in §1 shows is closest to a coin-flip (36.6% / 39.5% / 23.9%) — performance indicators are not marginal at all: **shots_on_target_diff** alone adds **+0.145 macro F1** over the scoreline, improving on *every single fold*. When a team is already leading or trailing, the scoreline has already settled the outcome so completely that adding a performance feature changes nothing measurable.

Practical reading: H5's "no survivors" verdict in §7 is an average over all match states, and it hides this. Performance indicators are not useless — they are useless *exactly where the scoreline has already answered the question*, and become the only real signal available in the one state where it hasn't.

9. §13 Extension — does the answer change in knockout football?

A secondary comparison, run on 250 StatsBomb tournament matches (FIFA World Cup, UEFA Euro, UEFA Women's Euro; group and knockout stages tagged directly from source, not inferred). Model A and Model C refit **separately within each stage** via grouped CV, so no fold leaks across competitions.

Stage	n	Model A (best)	Model C (best)
League (2015/16, 4 leagues)	1,517	0.6043	0.5626
Tournament group stage	178	0.4700	0.4988
Tournament knockout stage	94	0.5218	0.5654

Half-time-lead survival by format (does a lead hold to full time?):

Stage	Hold rate	n leads
League	79.6%	515
Tournament group	68.1%	47
Tournament knockout	90.9%	22

$\chi^2 = 5.39$, $p = 0.068$ — not significant at this sample size, but a real spread (68% to 91%) worth flagging for a larger follow-up.

The finding that matters: Model A drops **8.25 points** of macro F1 from league to knockout football (0.604 → 0.522) — the scoreline predicts knockout outcomes distinctly worse. Model C, over the same shift, barely moves (0.563 → 0.565) — **performance indicators close almost the entire gap**. The same-tournament control (group vs. knockout, format isolated from level of competition) shows the identical shape: A rises 0.470→0.522, C rises further, 0.499→0.565.

Verdict: this is exactly the pattern the project's title asks about — more room for "beyond the scoreline" signal exists precisely where the scoreline itself is weakest. It did not appear anywhere in league football (§5–§7 above); it appears clearly in knockout football. Caveat: knockout and group samples are in the low hundreds, so this is a lead for further work, not a result on the same footing as H1–H5.

10. Master result matrix

Hypothesis	Question	Primary evidence	Verdict
H1	Is the scoreline a strong predictor?	Model A macro F1 0.6073 vs baseline 0.2020, consistent across all 6 algorithms	Supported

Hypothesis	Question	Primary evidence	Verdict
H2	Is there signal in performance alone?	Model B macro F1 0.5219, above baseline but 0.0854 below Model A	Partially supported
H3	Does performance add to the scoreline?	Δ macro F1 -0.0125 ; McNemar significant 0/6 (CV); confirmed on frozen holdout	Not supported
H4	Which indicators add almost nothing?	Permutation vs. ablation agreement $r=0.918$; 2 indicators flagged by both	Supported — the scoreline dominates every performance feature
H5	Which indicators survive beyond the scoreline?	0 of 21 clear the CI + majority-of-folds bar	Not supported — but see §8: strongly supported <i>within the Drawing state specifically</i>
§13 (extension)	Does format change the answer?	Model A drops 8.3pts league→knockout; Model C nearly flat	Secondary finding, small sample
§14 (extension)	Does match state change the answer?	shots_on_target_diff: $+0.145$ macro F1, 11/11 folds, in Drawing state only	Strong secondary finding

11. Known limitations

- **One season, four leagues.** StatsBomb's complete multi-league coverage is limited to 2015/16; the proposal's season-by-season expanding-window CV (§8) was implemented **by match-week within the season** instead, since no multi-season complete coverage exists for any league in the archive.
- **The frozen holdout is a late-season block**, not a random sample — motivation shifts once titles and relegation are settled, so it is an honest future test, not an unbiased one.
- **§13's tournament sample (250 of 427 available matches) was ingested under a tightened time budget**; the group/knockout comparison is directionally clear but should be treated as preliminary given sample size.
- **Collinearity limits what ablation alone can prove**: passing and possession features are highly correlated with each other (mean $|r|$ across the block, computed in the notebook §3.1), so a near-zero ablation delta means "replaceable by a correlated feature," not necessarily "uninformative" — which is exactly why H4 and H5 are read together rather than singly.
- **Pre-match context (team strength, form, fixture congestion) is out of scope** — this dataset measures what the state of play adds, not what a complete pre-match model would.

12. Reproducing every number in this document

```
cd Final_hypo/src
python ingest.py          # rebuild the league dataset from StatsBomb events
python ingest.py tournaments # rebuild the tournament dataset (optional, §13)
```



```
only)
python run_all.py          # features -> train -> H4/H5 -> frozen holdout ->
$13 -> notebook
```

Full deliverable: [Final_hypo/notebooks/beyond_the_scoreline_final.ipynb](#) (52 cells, markdown narrative + code + graphs, every verdict computed live from the CSVs in [Final_hypo/results/](#) rather than hard-coded).